VICTORY 2027

The Nervous System



LECTURE-04 BIOLOGY

By : Upasana Ma'am

Topics to be Covered

1. Spinal Cord
2. PNS, Reflex arc

Jmp. Que.


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上



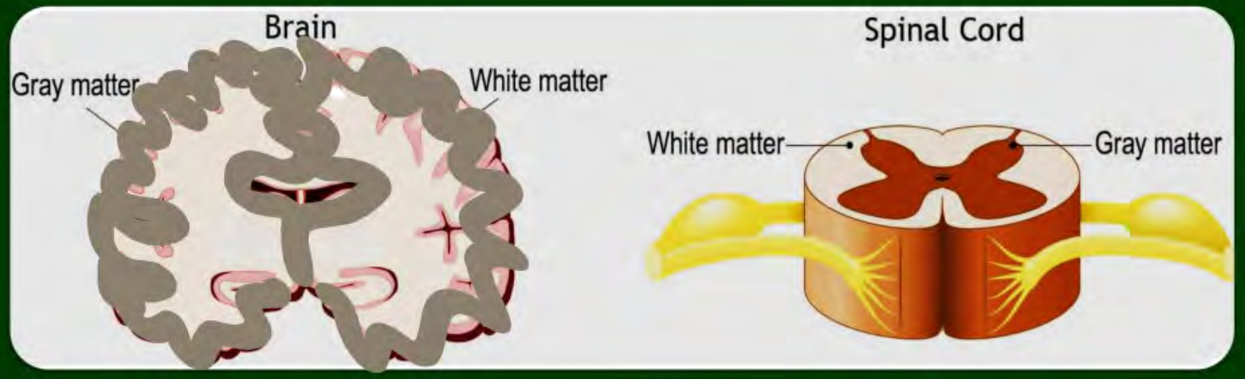
GREY MATTER

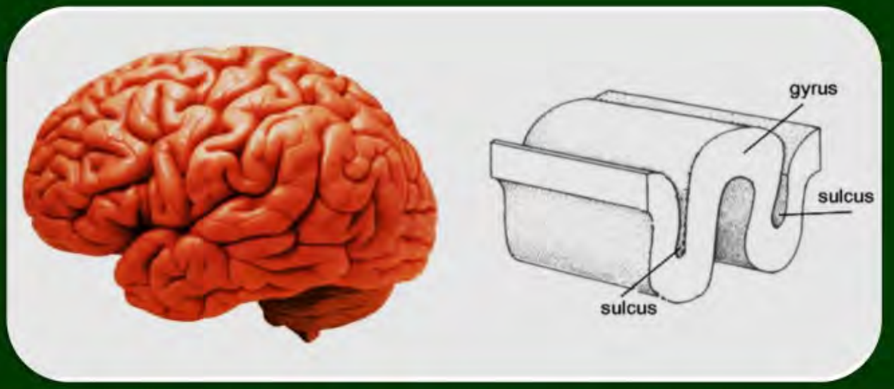
- Found in outer cortex region of cerebrum.
- Contains cell bodies of the neurons.
- Appears greyish in colour.
- Folded to form the convolutions.
- The folds are called gyri and the grooves are called sulci (creases).

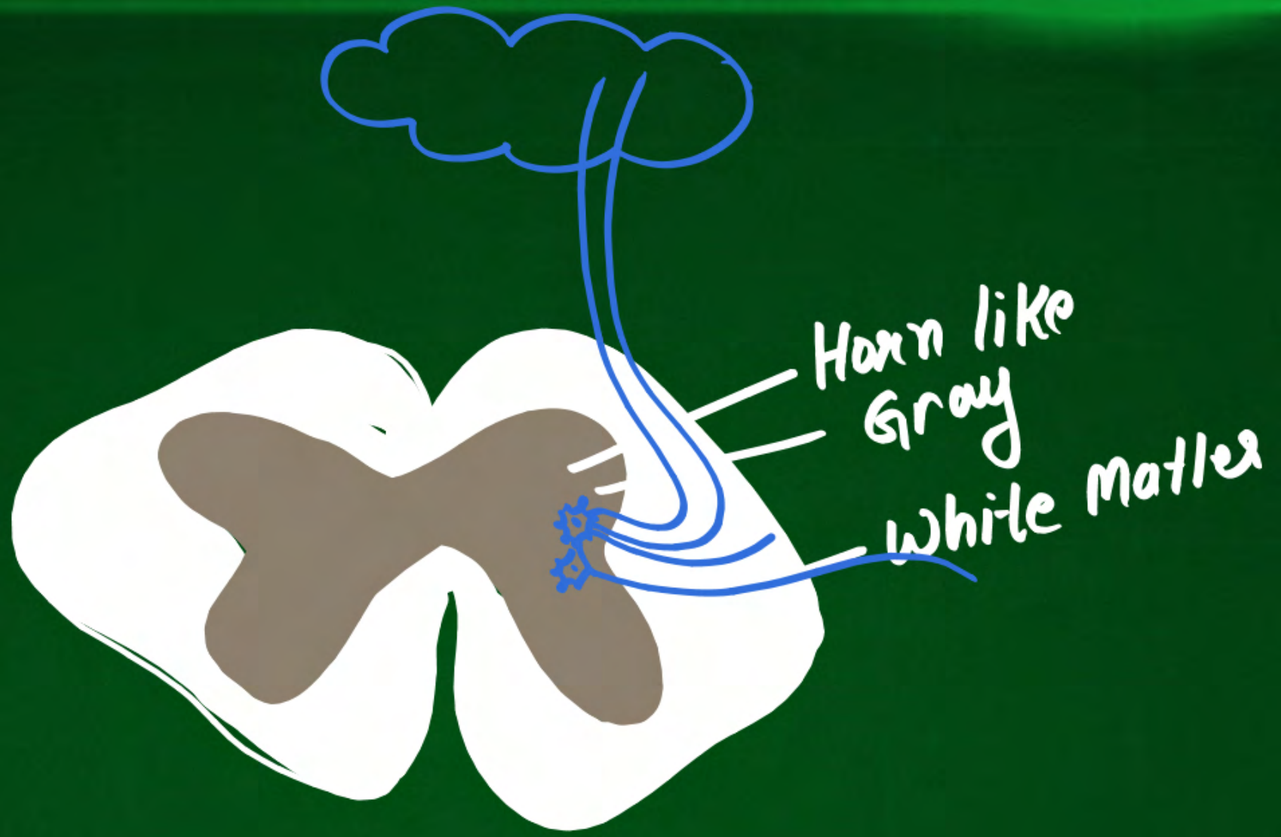
WHITE MATTER

- Inner portion of the cerebrum. (medulla)
- Contains mainly the axons (nerve fibres) of the neurons.

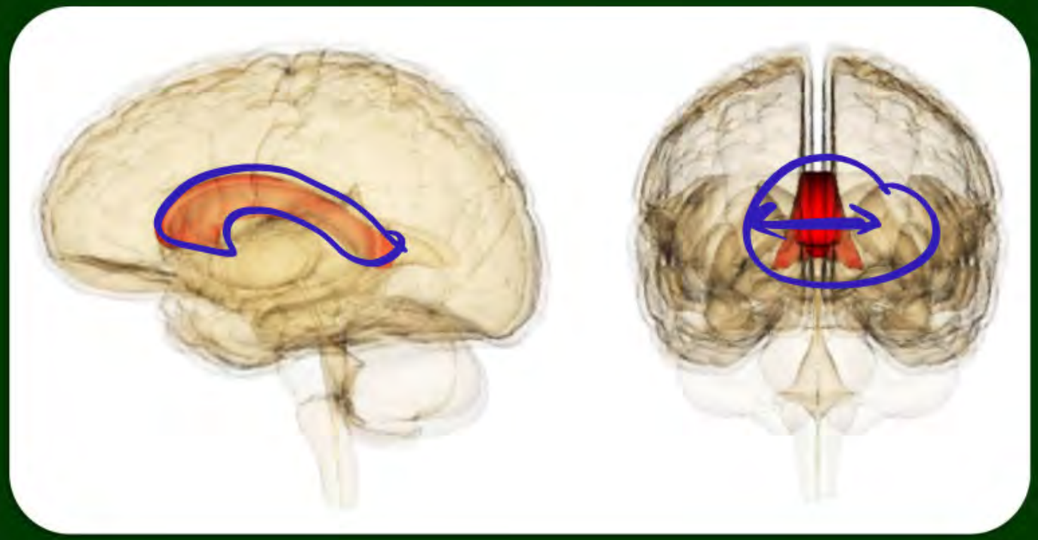
Spinal Cord






Horn like
Gray
White matter

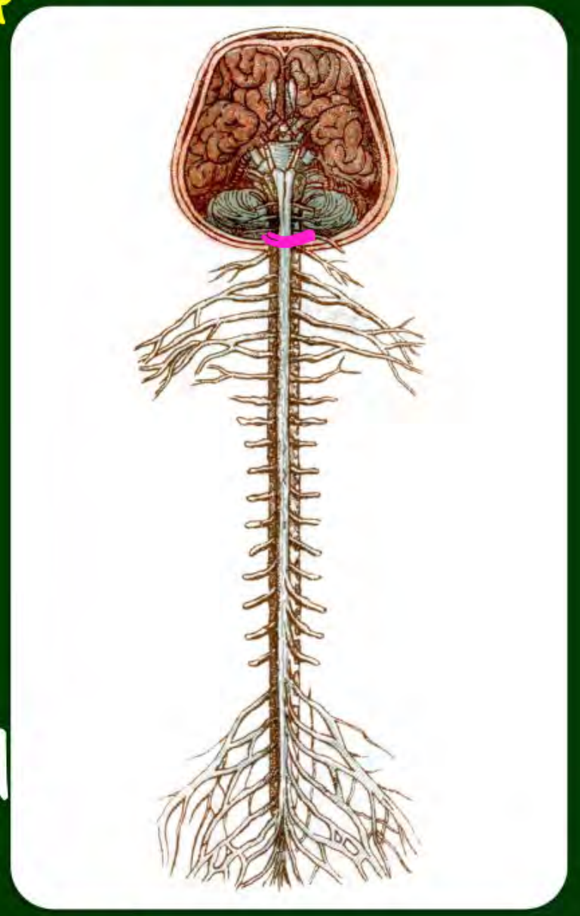
- **Corpus callosum** (hard body) is a sheet of fibres connecting the two cerebral hemispheres.
- **Its function is to transfer information from one hemisphere to the other.**

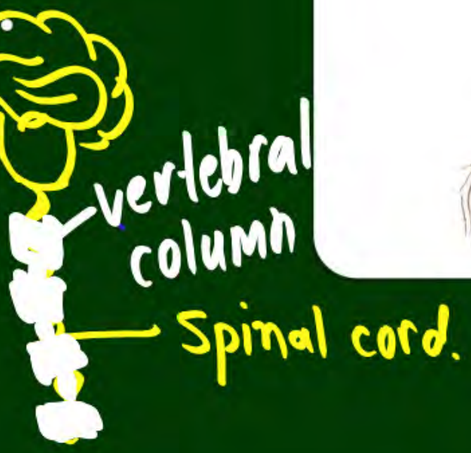


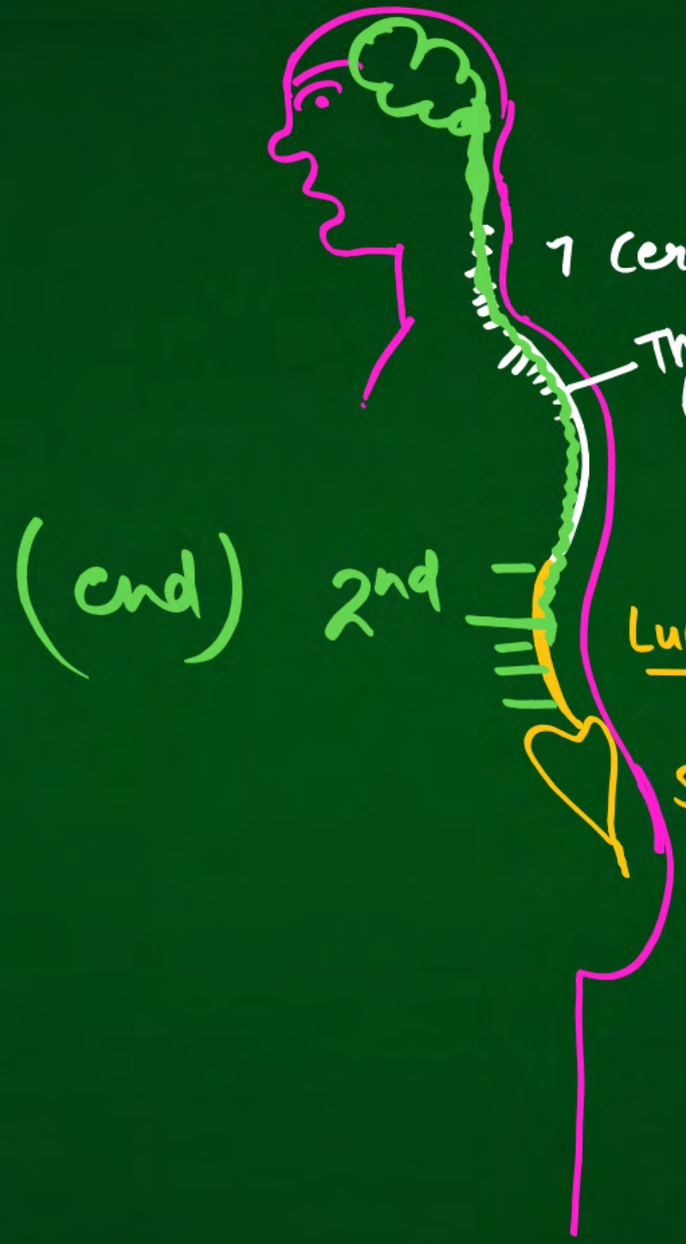
Spinal Cord

- Spinal cord is a cylindrical structure protected by bony cage called vertebral column.
- Begins in continuation with medulla and extends downwards till second lumbar vertebra, lies within the neural canal of the vertebrae.
- Central canal is present in the centre and is continuous with ventricles.
- Filled with CSF to absorb shock and form medium to exchange food, waste and gases.
- Also covered with 3 meninges.



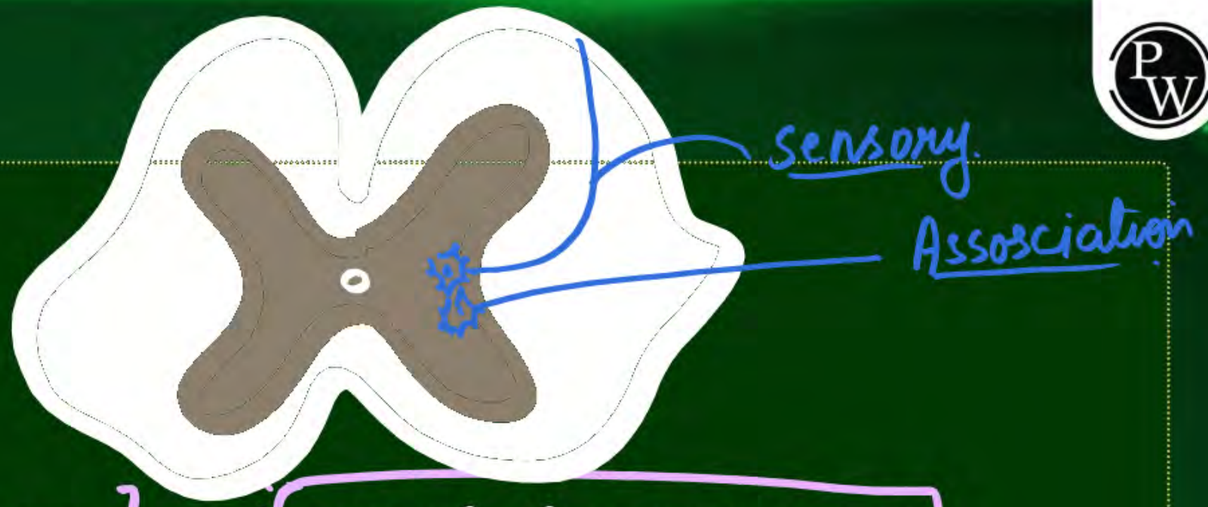






[26 Vertebrae]

Spinal Cord



Gray matter

- **Lies inside**
- Contains Cell bodies of afferent neurons and association neurons.

White matter

- *Lies outside*
- *Contains axons longitudinally to and from the brain and other sides.*
- Spinal cord is also surrounded by membranes called meninges.
- As many as 31 pairs of nerves arise from the spinal cord.

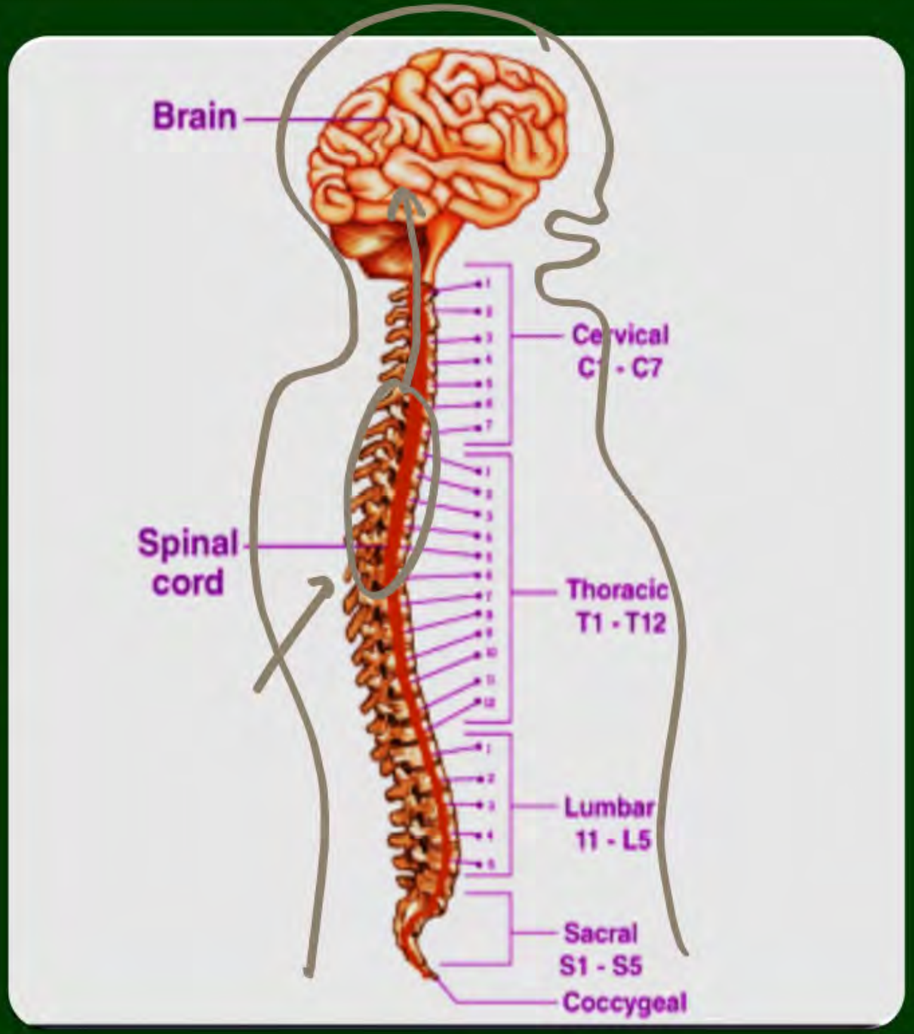
Functions of the spinal cord :

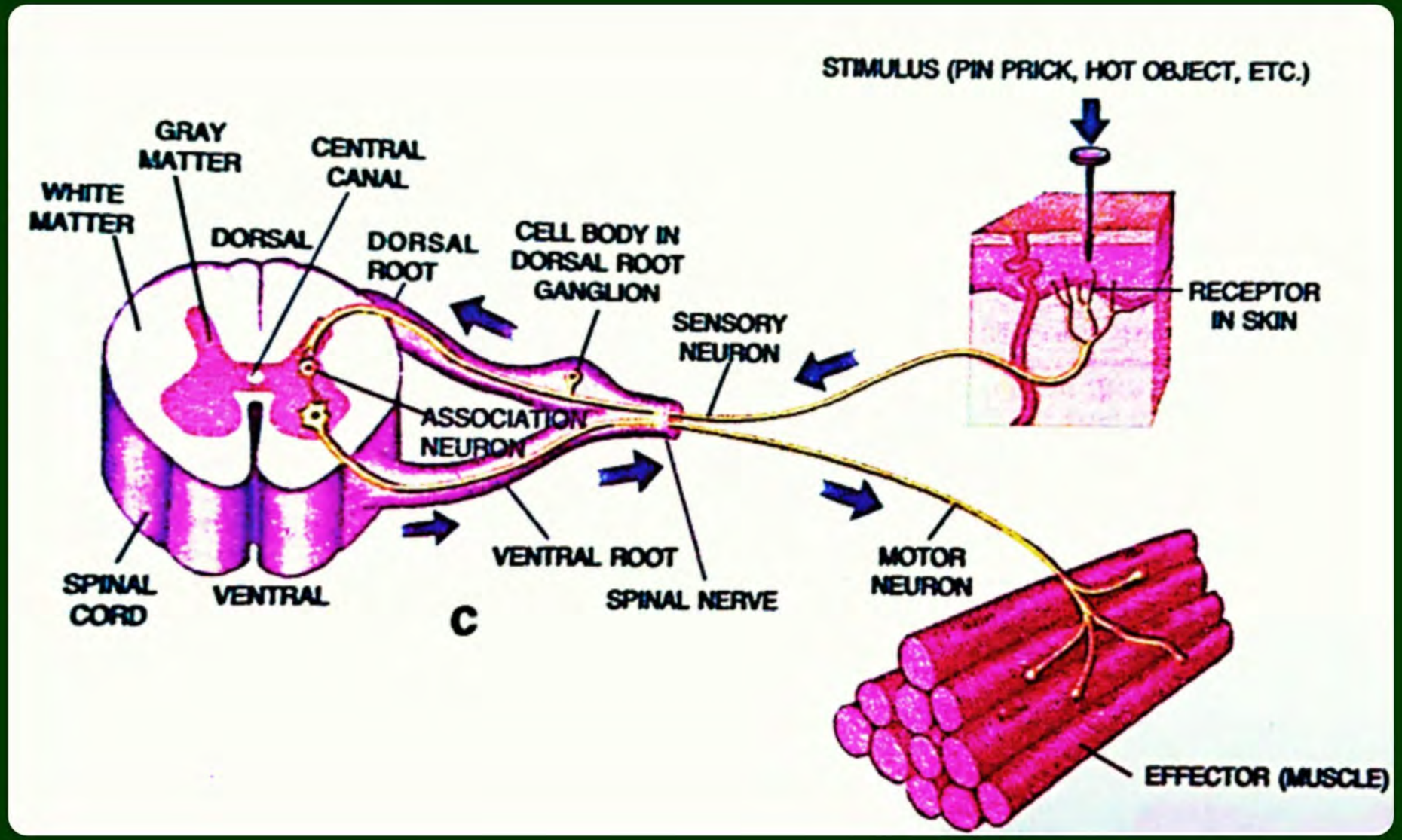
(i) Reflexes below the neck.

(ii) Conducts sensory impulses from the skin and muscles to the brain.

(iii) Conducts motor responses from the brain to muscles of the trunk and limbs.

NOTE → Brain has 4 Ventricles filled with fluid




STIMULUS (PIN PRICK, HOT OBJECT, ETC.)
GRAY MATTER
CENTRAL CANAL
WHITE MATTER
DORSAL
DORSAL ROOT
CELL BODY IN DORSAL ROOT GANGLION
SENSORY NEURON
RECEPTOR IN SKIN
ASSOCIATION NEURON
VENTRAL ROOT
SPINAL NERVE
MOTOR NEURON
SPINAL CORD
VENTRAL
C
EFFECTOR (MUSCLE)

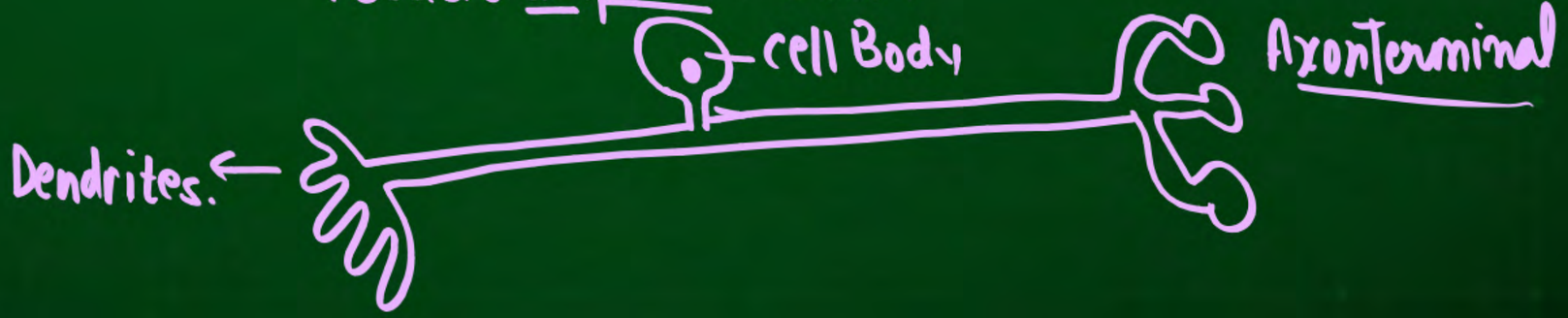
Reflex Action.

'Types of Neurons'

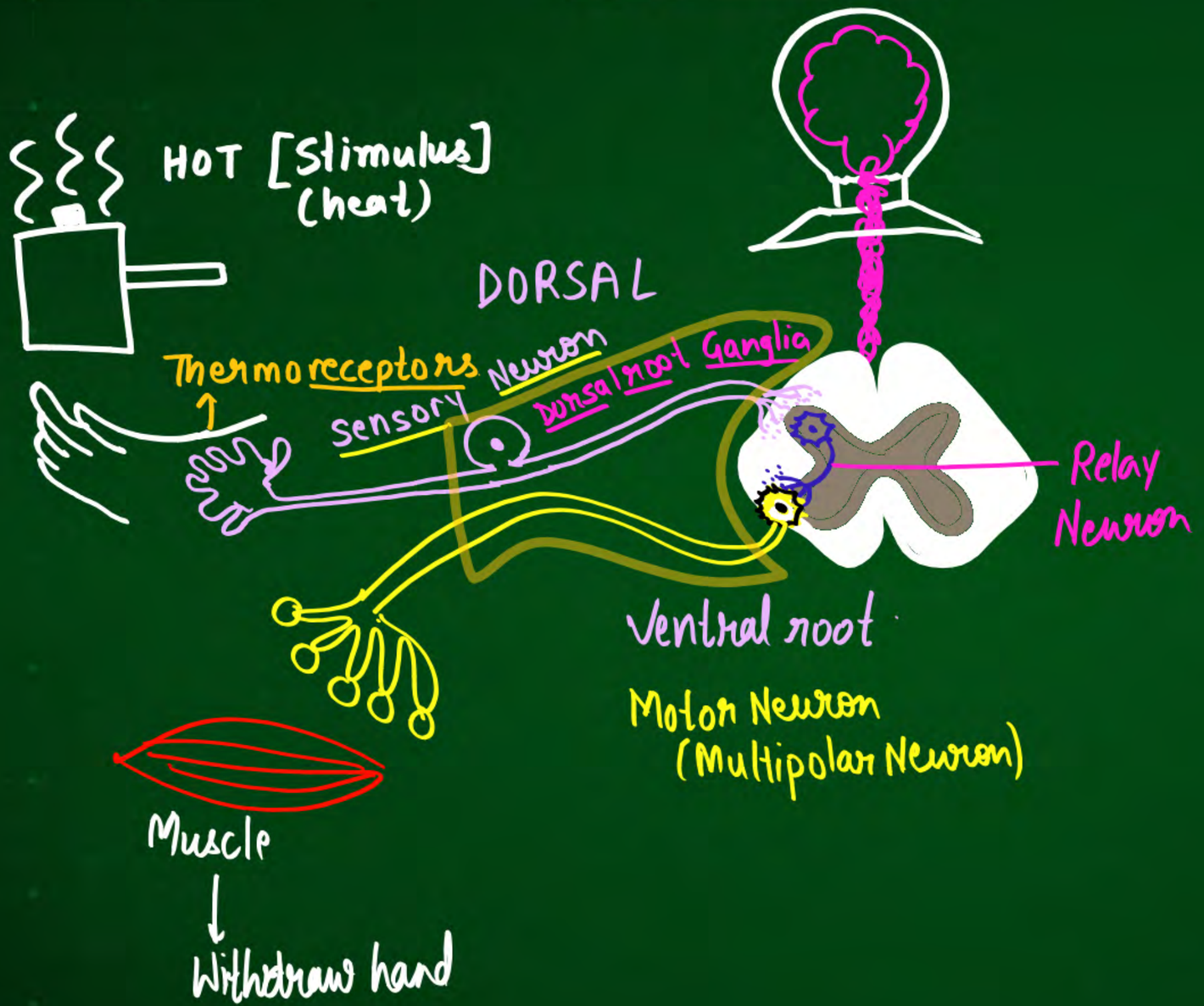
'Multipolar Neuron'



Pseudo unipolar Neuron



Reflex Action.



SPINAL CORD
(MIXED NERVES)

Peripheral Nervous System (PNS)



Body

Somatic Nervous System (Voluntary actions)

- Cranial nerves (12 pairs)
- Spinal nerves (31 pairs)

Autonomic Nervous System (ANS)

- Sympathetic
- (Parasympathetic Nervous System)

(Stimulatory)

Accelerates heart beat, dilates, pupil, constricts blood vessels, releases sugar from liver

(Calming)

Restores normal body conditions

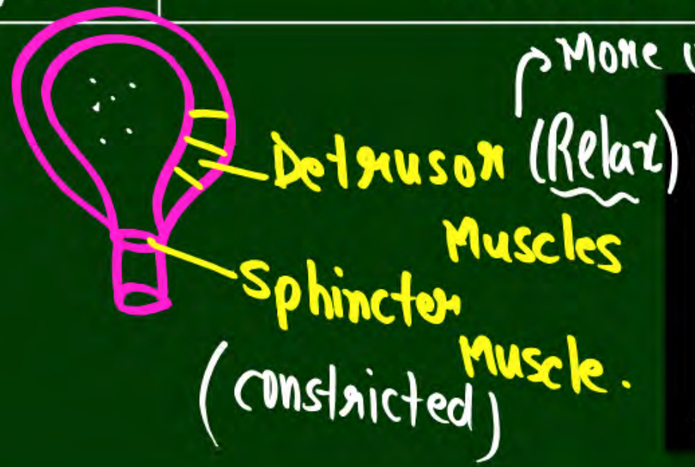
Effects of the Two Parts of Autonomic Nervous System (ANS)

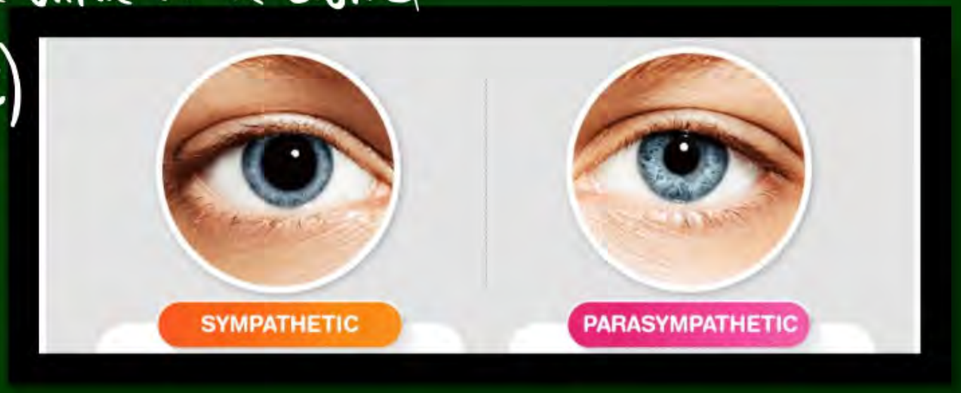
S. No.	Organs	Sympathetic system	Parasympathetic system
1.	Heart	Accelerates heart beat	Retards heart beat
2.	Blood vessels	Constricts all blood vessels except coronary vessels which are dilated	Dilates all blood vessels except coronary vessels which are constricted



S. No.	Organs	Sympathetic system	Parasympathetic system
3.	Lungs	Dilates bronchi and bronchioles	Constricts bronchi and bronchioles
4.	Intestines	Peristalsis decreased	Peristalsis increased

S. No.	Organs	Sympathetic system	Parasympathetic system
5.	Urinary bladder	Sphincter contraction, Muscle relaxed	Sphincter relaxation, muscle contraction (feeling to urinate)
6.	Pupil of eye	Dilation	Constriction





S. No.	Organs	Sympathetic system	Parasympathetic system
7.	Salivary glands	Inhibits secretion of saliva ↓ (dryness of mouth)	Stimulates secretion of saliva
8.	Lacrimal (tear) glands	Inhibits secretion ↓	Stimulates secretion

S. No.	Organs	Sympathetic system	Parasympathetic system
9.	Arrector (or erector) pili (hair) muscles of skin)	Stimulates contraction (hair raised)	Relaxes (hair flattened)
10.	Body (as a whole)	Prepares body for action	Prepares body for relaxation.



Emotions and Autonomic Nervous System

- Autonomic nervous system is strongly influenced by emotions such as grief, anger, fear, sexual stimulation, etc.
- High blood pressure, stomach ulcers and some other troubles may arise due to long continued emotional stress.

Peripheral Nervous System (PNS)

Peripheral nervous system (PNS) includes the nerves that emerge from and enter into the brain and spinal cord.

A. **Somatic nervous system (SNS)** conveys information to skeletal (voluntary) muscles.

Cranial nerves:

- Emerge from the brain.
- 12 pairs of cranial nerves.
- **Sensory -** olfactory, optic and auditory.
- **Motor -** going to eye muscle.
- **Mixed -** going to and coming from face and tongue.

Spinal nerves

- Emerge from the spinal cord 31 pairs of spinal nerves
- Neck(8),thorax (12), lumbar region (5), sacral (5), coccygeal region (1)

Peripheral Nervous System (PNS)

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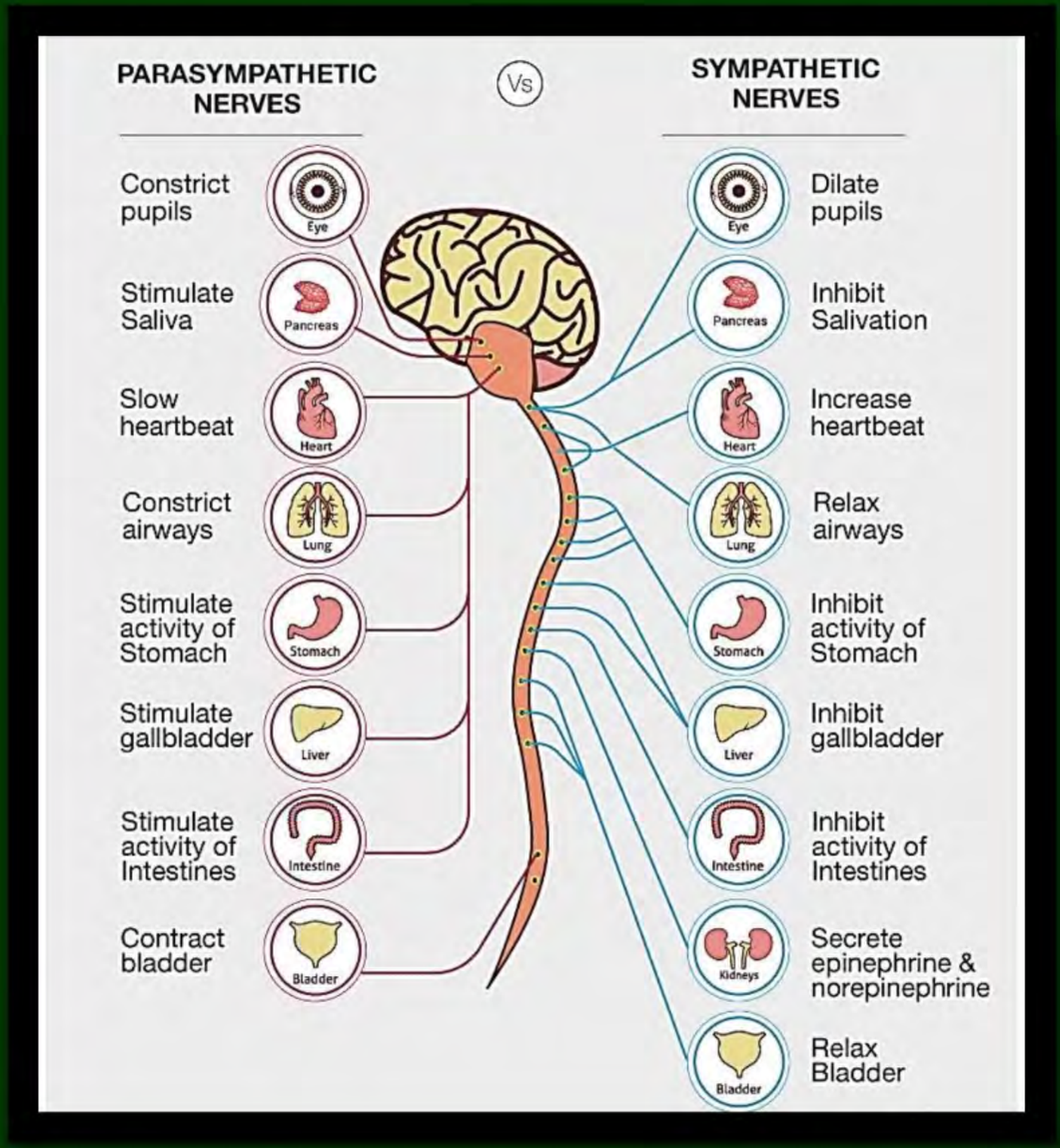
A. Autonomic nervous system (SNS).

B. Consists of a pair of chains of nerves and ganglia on either side of the backbone.

C. Controls involuntary actions.

Parasympathetic nervous sytem
- It is located at 2 places
- Anteriorly in head and neck
- Posteriorly in sacral region

Sympathetic nervous system:
- Nerves arise from the spinal cord between Neck and waist region Stimulated by adrenaline hormone.


PARASYMPATHETIC NERVES
Vs
SYMPATHETIC NERVES
Constrict pupils
Eye
Stimulate Saliva
Pancreas
Slow heartbeat
Heart
Constrict airways
Lung
Stimulate activity of Stomach
Stomach
Stimulate gallbladder
Liver
Stimulate activity of Intestines
Intestine
Contract bladder
Bladder
Dilate pupils
Eye
Inhibit Salivation
Pancreas
Increase heartbeat
Heart
Relax airways
Lung
Inhibit activity of Stomach
Stomach
Inhibit gallbladder
Liver
Inhibit activity of Intestines
Intestine
Secrete epinephrine & norepinephrine
Kidneys
Relax Bladder
Bladder

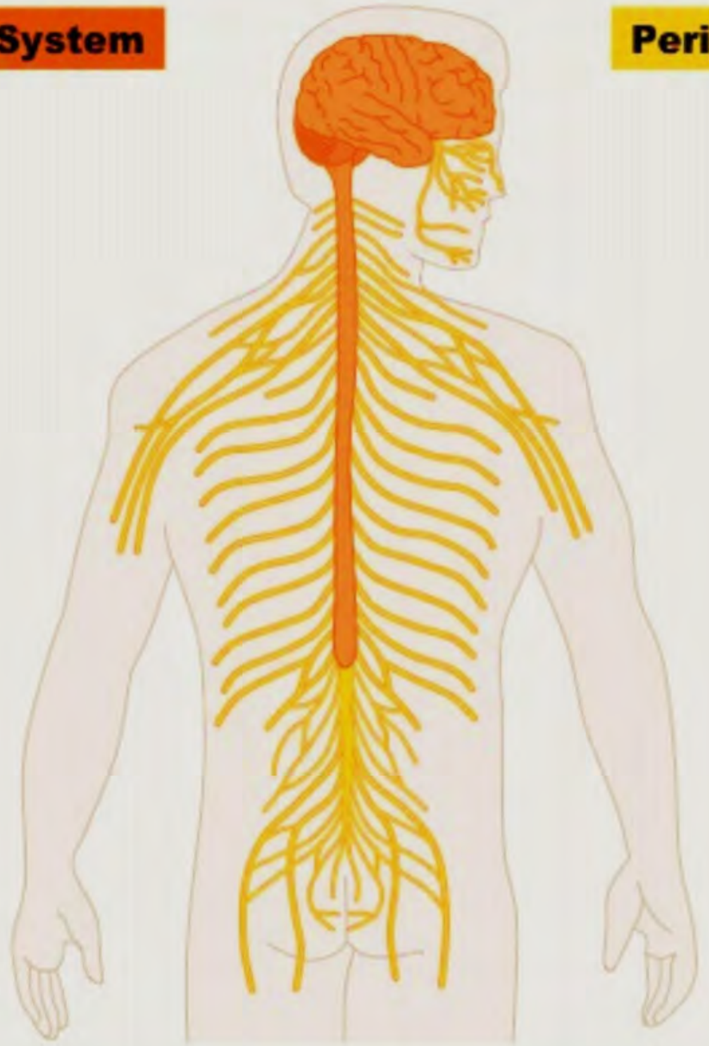
Central Nervous System

Composed of:

- Brain
- Spinal cord

Contains:

- Relay neurons (interneurons)



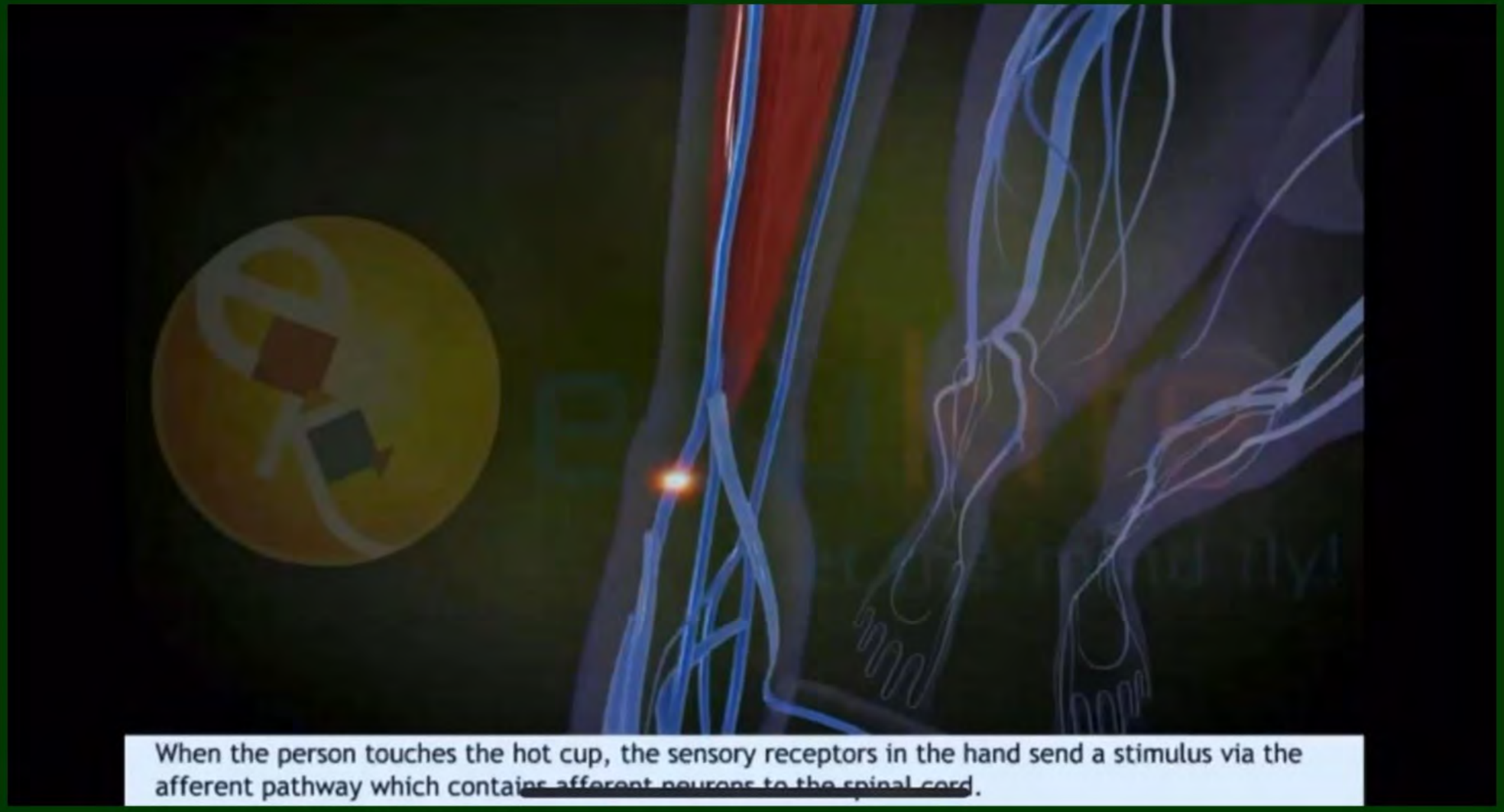
Peripheral Nervous System

Composed of:

- Cranial nerves
- Spinal nerves
- Peripheral nerves

Contains:

- Sensory neurons
- Motor neurons


When the person touches the hot cup, the sensory receptors in the hand send a stimulus via the
afferent pathway which contains afferent neurons to the spinal cord.

REFLEX

- The term "reflex" comes from the Latin word "reflexus" meaning reflected or directed back.
- **There are two types of actions which occur in our body:**
- Voluntary (performed consciously) and
- Involuntary (performed unconsciously).

Examples:

1. Voluntary actions

- To watch some programme on TV.
- Pick up an apple and eat it.
- Know the time, you raise your arm.

2. Involuntary actions

- It is an automatic/quick/immediate, involuntary action in the body brought about by a stimulus.
- Some particle falls into your eye and there is immediate flushing of tears to wash out the particle (glandular secretion).
- Instantaneous withdrawal of hand when it accidentally touches a hot iron (muscular movement).

A reflex arc may be represented as follows :

❖ Stimulus → receptor in the sense organ → afferent (sensory) nerve fibre → CNS (spinal cord/brain) → efferent (motor) nerve fibre → muscle (to contract) / gland (to secrete).)

Types of reflexes:

1. **Natural (inborn) reflex:**

- Is one in which no previous experience or learning is required.
- These reflexes are inborn, i.e., inherited from the parents.

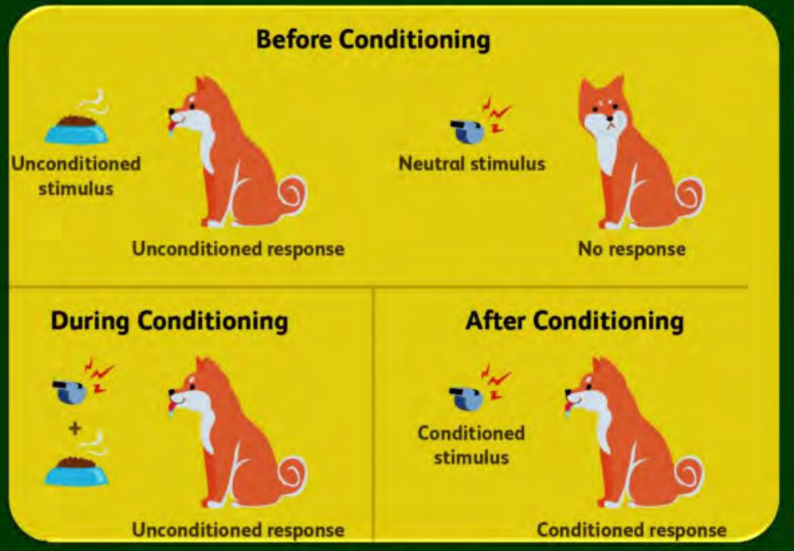
Examples :

- **Blinking, coughing, sneezing :**
- These are protective.
- Salivation, swallowing, peristalsis : These provide functional efficiency.

2. **Conditioned (or acquired) reflex:**

- Is one which develops during lifetime due to experience or learning.
- For example, what you experience simply at the sight or by the smell of a familiar tasty food resulting in watering (salivation) of your mouth.





- **Response:** The change in an organism resulting due to stimulus.
- **Impulse:** A wave of irritability (electrical disturbance that sweeps over the nerve cell).
- **Receptors:** the specialised epithelial cells which, on receiving the stimulus, set up waves of impulses towards the central nervous system. e.g. photoreceptors (sight), phonoreceptors (sound), thermoreceptors (temperature), olfactory receptors (smell) and so on present in our sense organs.
- **Effectors:** muscles or glands which, on receiving the impulse from the brain or spinal cord, contract or secrete substances.

Grey matter in spinal cord lies –

A) Inside ✓

B) Outside.

Sensory impulse in reflex action enters ___

A) Dorsally

B) Ventrally

QUESTION

Corpus callosum connects

A. Two cerebral hemispheres

B. Diencephalon and mid brain

C. Pons and medulla

D. Medulla and spinal cord

QUESTION

_____ are the neurons carrying impulses away from the central nervous system.

A. Efferent nerves (Motor)

B. Afferent nerves (आने) Sensory

C. Extensors

D. Sensory nerves

CNS → Effector.

QUESTION

Myelin Sheath Last Year 'Q' (False)

Assertion (A): Arachnoid layer helps in speeding up the transmission of nerve impulses along the axons.

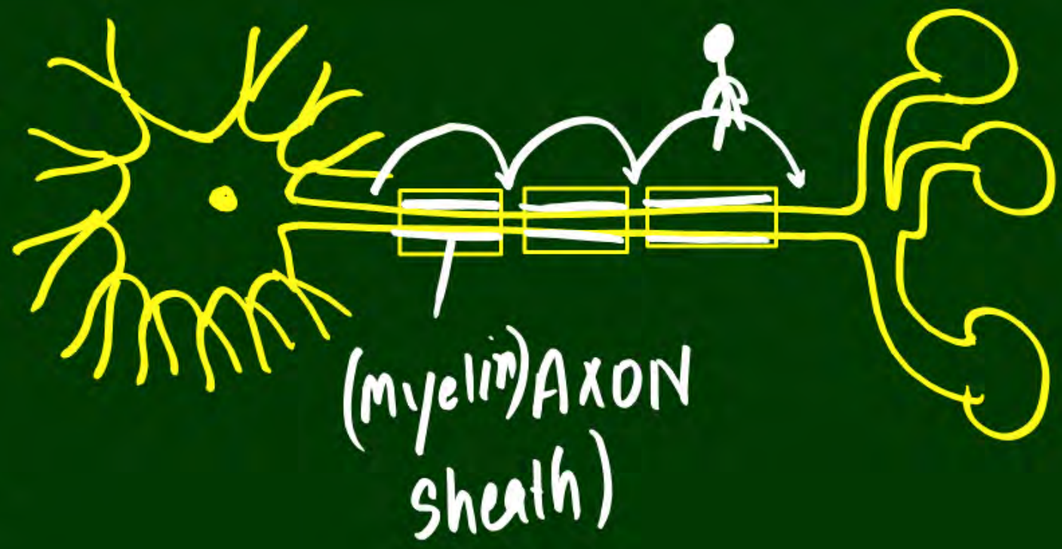
Reason (R): Axon endings at the synapse secrete acetylene. 'acetylcholine'

A. (A) is true and (R) is false.

B. (A) is false and (R) is true.

C. Both (A) and (R) are true.

D. Both (A) and (R) are false.



QUESTION

'Competency Based'

Vincet's day begins with a cup of piping hot coffee. Caffeine is a stimulant present in coffee which increases the circulation of Adrenaline in the body. Vineet feels refreshed, alert and focussed. [3]

(a) Name the organ that is stimulated by caffeine which makes Vineet alertand focussed. Cerebrum

(b) Give the collective term for the protective membranes of the organ mentioned in (a) Meninges

(c) Which part of the autonomic nervous system increases the secretion of Adrenaline? Sympathetic

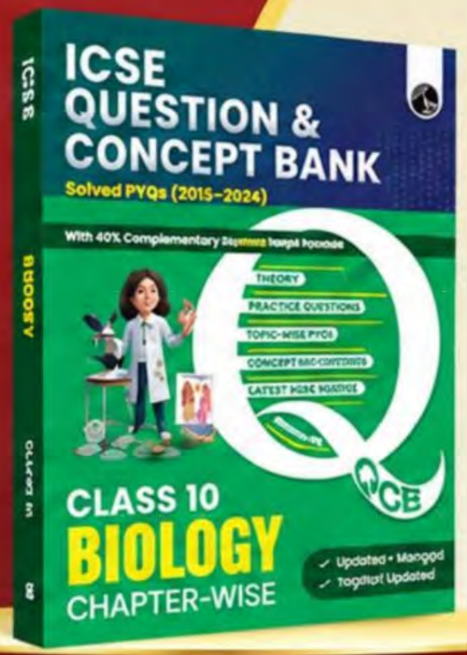


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